

Phenotype-Conditional Diffusion for Single-Cell Gene Expression Generation

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1. Brief Description of the Research Project

This project develops a **conditional diffusion model** that generates single-cell gene expression profiles conditioned on cell-type labels. The work transfers two ideas from vision generative modeling — denoising diffusion probabilistic models (DDPM) and classifier-free guidance (CFG) — into a non-image biological modality. Given a target cell-type label c (e.g., T cell, B cell, Monocyte), the model produces a synthetic expression vector $x \in \mathbb{R}^G$ over G highly variable genes, capturing both the marginal expression distribution and cell-type-specific marker gene patterns. We use the public PBMC 3k dataset (~2,700 cells, 8 annotated cell types) as the target distribution, and evaluate the generated samples by distributional metrics (MMD, Wasserstein distance), marker-gene validation, and UMAP cluster separation.

2. Motivation & Problem Statement

Vision-domain generative models (DDPM, latent diffusion, classifier-free guidance) have set the state of the art for *conditional* synthesis of structured data, but most biology-facing generative work on single-cell RNA-seq (scRNA-seq) has historically relied on VAE-based models such as scVI [1] and scVAE [2], which are optimized primarily for representation learning and clustering rather than high-fidelity conditional generation. Two specific gaps motivate this project. **(i)** scRNA-seq experiments are expensive, sparse, and frequently lack samples of rare cell types; high-fidelity conditional generators can serve as data augmentation for downstream classification, perturbation analysis, and rare-population recovery [3, 4]. **(ii)** The mechanics of vision-style conditional diffusion — Gaussian noising over a continuous vector and a learned denoiser conditioned on a class embedding — transfer naturally to scRNA-seq expression vectors once preprocessing (normalization, log-transform, HVG selection) is applied. Recent work such as scDiffusion [3] and LapDDPM [4] confirms this direction is viable, but the architectures and training pipelines are substantially heavier than necessary for a course-scale study.

The **problem statement** of this project is therefore: *can a minimal 1D DDPM with classifier-free guidance, trained directly on log-normalized expression vectors, generate cell-type-conditional scRNA-seq samples that are statistically and biologically faithful to the PBMC reference distribution?* The project does not aim to outperform domain-specific state-of-the-art; it aims to isolate the core vision-generative-model mechanism and verify it transfers to a biological modality under a controlled, reproducible setup.

3. Proposed Method / Technical Details

Dataset & preprocessing. We use `scanpy.datasets.pbmc3k()` [5, 6] which provides 2,700 PBMCs from a healthy donor [7]. The standard scanpy pipeline is applied: filter cells (min 200 genes) and genes (min 3 cells), normalize total counts to $1e4$, log1p-transform, and retain the top $G = 1,000$ highly variable genes. Cell-type labels (8 classes) are obtained from the processed version `pbmc3k_processed()`. The final input is a matrix $X \in \mathbb{R}^{2638 \times 1000}$ with per-cell label $c \in \{0, \dots, 7\}$.

Generative model. We train a 1D conditional DDPM [8] directly on the log-normalized expression vector x_0 . The forward process follows the standard variance-preserving formulation $q(x_t|x_0) = \mathcal{N}(\sqrt{\bar{\alpha}_t} x_0, (1-\bar{\alpha}_t) I)$ with a linear β -schedule and $T = 1000$ steps. The denoiser $\varepsilon_\theta(x_t, t, c)$ is a residual MLP backbone (4 blocks, hidden 512, SiLU + LayerNorm), with sinusoidal time embedding and a learned class-label embedding added at every block. Classifier-free guidance [9] is applied by randomly dropping the label (replacing it with a null token) with probability $p = 0.1$ during training; at sampling time, guided scores are computed as $\tilde{\varepsilon} = (1+w) \cdot \varepsilon_\theta(x_t, t, c) - w \cdot \varepsilon_\theta(x_t, t, \emptyset)$ with guidance scale $w \in \{0, 1, 3, 5\}$. We train for 200 epochs with AdamW (lr = $1e-4$, batch 256) and report results on the best validation MMD checkpoint.

Sampling. We use DDIM [10] sampling with 50 steps for efficiency. For each target class c , we draw $N = 1,000$ synthetic cells and concatenate them with the real test set for evaluation.

Evaluation. Three complementary metrics are reported per class and aggregated:

- **Distributional fidelity:** Maximum Mean Discrepancy (MMD, RBF kernel) and 2-Wasserstein distance between real and generated samples in both ambient and PCA-reduced (50-dim) space.
- **Biological validity:** expression level of canonical marker genes per class (e.g., CD3D for T cells, MS4A1 for B cells, LYZ for monocytes) compared against real cells via two-sample KS test.
- **Geometric structure:** joint UMAP of real + generated cells; we check whether generated cells co-cluster with the correct real cell-type cluster.

Baselines. We compare against (i) a conditional VAE trained on the same input space, and (ii) per-class Gaussian sampling on the PCA-reduced space, to confirm that the diffusion component contributes beyond simple latent Gaussian modeling.

Implementation & reproducibility. The pipeline is implemented in PyTorch with scanpy and anndata for data handling. Code, the preprocessed AnnData file, training configs, and all evaluation scripts will be released in a single repository. The full training run on PBMC 3k is expected to complete in under 2 hours on a single NVIDIA RTX 3090, making the entire study reproducible end-to-end.

4. References

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